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Development of a multifunctional neoglycoside auxiliary for applications in glycomics research†

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A novel, multifunctional, tetrazine-containing neoglycoside auxiliary has been synthesized in three steps and 28% overall yield. The oxyamine was conjugated with unprotected carbohydrates under aqueous conditions (pH = 4.7), with DMF as a cosolvent, to provide neoglycosides in yields ranging between 51% and 68%. This auxiliary displayed broad advantages in the isolation and purification of complex carbohydrate mixtures, compatibility during extension by glycosyltransferases, and direct conjugation to chemical probes. Furthermore, the auxiliary can be removed in 96% yield under acidic conditions (0.25% TFA in H_2O) that leave glycosidic linkages intact. Thereby, the tetrazine-containing neoglycoside auxiliary can serve to facilitate future glycomics investigations.

Glycans, and more frequently the glycoconjugates of proteins and lipids, are implicated in virtually all important biological processes.[1] These conjugated glycans can be recognized by glycan-binding proteins (GBPs). The interactions between glycans and GBPs play crucial roles in biological recognition processes such as cell signalling, protein folding, cell adhesion, host–pathogen interactions, *etc.*[1,2] The structural complexities and diverse biological functions of glycans necessitate reliable and efficient techniques to conduct glycomic studies. The utilization of mass spectrometry allows the rapid determination of the connectivity of complex carbohydrates.[3] However, mass spectrometric investigations typically utilize separation techniques such as capillary electrophoresis or liquid chromatography to effectively analyze glycan mixtures, especially isomeric species.[4] The study of glycan-protein interactions can be interrogated using microarray technologies in which glycans are modified for the attachment to a surface and visualized by fluorescent probes.[5] Separation technologies and glycan array manufacturing regularly utilize sugars modified with organic moieties that can be hydrophobic, UV-active, or modifiable.

Chemoselective ligation reactions at the anomeric center of carbohydrates have been extensively developed during the past decade.[6] The attachment of an auxiliary to a reducing sugar facilitates examination of biological interactions through the introduction of fluorescent tags, immobilization on surface microarray, and conjugation to biomolecules.[7] A neoglycoside auxiliary (secondary oxyamine) is a privileged scaffold for glycan modification because it maximizes the likelihood of recognition by GBPs due to preservation of a cyclic heminal. Several elegant auxiliaries have been reported exhibiting compatibility with chemoenzymatic extension and/or highly efficient conjugation to surfaces for the production of glycan microarrays.[8] However, in order to utilize these probes in glycomic applications, further chemical modifications were required for conjugation to a probe. The purpose of this work is to develop a multifunctional auxiliary that can serve in a variety of glycomic applications and also enable direct bioorthogonal conjugation without chemical modification.

Inverse-electron-demand Diels–Alder (IEDDA) reactions between tetrazines and strained dienophiles such as *trans*-cyclooctene (TCO), cyclooctyne, and norbornene, have gained attention recently due to fast and highly chemoselective cycloadditions.[9] The rate constants of the IEDDA reaction between tetrazine and TCO can be as high as 10^6 M^{-1} s^{-1}.[10] The exceptional kinetic features have been widely used to label small molecules for visualization under physiological condition.[11] Recently, Sun *et al.* developed a microarray immobilization technique by coupling tetrazine-containing peptides onto TCO-functionalized slides in a rapid and chemoselective manner.[12] Therefore, tetrazine-containig neoglycoside auxiliary could be used for labeling and studying the biological functions of glycans *via* IEDDA reaction.

Herein, the synthesis of a multifunctional neoglycoside auxiliary and the applications of this auxiliary are reported. The hydrophobic, UV-active fragment of the auxiliary facilitates the isolation and purification of complex carbohydrate mix-

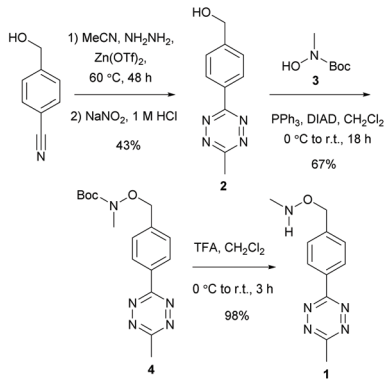
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tures by reverse-phase chromatography. The tetrazine auxiliary contains non-invasive functional groups to maximize compatibility with enzymes and cells. The neoglycoside linkage is stable over the pH range of most glycomic investigations yet quantitatively removable. Moreover, the tetrazine moiety on the auxiliary enabled direct conjugation to chemical probes without further modification.

The synthetic route for the production of the tetrazine auxiliary **1** is shown in Scheme 1. Commercially available 4-(hydroxymethyl)-benzonitrile is joined with acetonitrile and hydrazine in the presence of zinc triflate in a one-pot manner to form hydroxybenzyl tetrazine (**2**).[13] An extended reaction time of 48 h was necessary to improve the reported of 27% yield to a modest, but useful 43%. The benzylic alcohol of **2** was displaced with *N*-Boc-*N*′-methylhydroxylamine **3** using standard Mitsunobu conditions, and subsequently purified by silica gel chromatography to obtain a Boc-protected tetrazine-oxyamine (**4**). Finally, acidic removal of the *tert*-butyl carbamate protecting group with trifluoroacetic acid provides the tetrazine auxiliary **1** in high yield and purity after simple extraction from a basic aqueous solution. The UV-vis absorption spectrum of the tetrazine auxiliary **1** revealed two characteristic maxima absorption wavelengths at 270 nm and at 540 nm (see ESI†). The UV-vis absorption property of auxiliary **1** provides the ability to monitor the reaction at either UV or visible region to avoid the interference of other compounds in the reaction mixture.

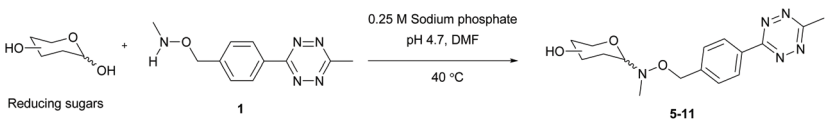


Scheme 1 Synthesis of multifunctional tetrazine auxiliary (**1**).

With the auxiliary **1** in hand, a glycoconjugate library was created by derivatization of a variety of reducing sugars (Table 1). The rate and efficiency of the oxyamine ligation is heavily dependent on the concentration of the saccharide and oxyamine, as well as the solvent and acid catalyst.[14] The most efficient and general conditions utilized a sodium phosphate buffer (pH = 4.7) although other buffers (NH_4OAc and NaOAc) were found to be viable. The rate of ligation was significantly improved by utilizing *N*,*N*-dimethylformamide (DMF) as a cosolvent to improve the solubility of the auxiliary in the reaction medium. The optimal stoichiometry for the reaction was evaluated to maximize the efficiency of the saccharide and auxiliary. Two complementary reactions (Table 1, entries 1 and 2) demonstrated that the utilization of a five-fold molar excess of either the sugar or the auxiliary resulted in an acceptable yield of the conjugate. All subsequent entries in Table 1 utilized an excess of the reducing sugars due to the substrates being readily available and inexpensive. The conjugation of seven reducing sugars were carried out under the established conditions and purified by reverse-phase chromatography on C18 functionalized silica gel. Hexoses, pentoses, disaccharides, and amido-glycans were examined and provided the expected products with yields ranging between 46% and 68%.

The conjugated products were characterized by NMR and mass spectrometry. The stereochemistry at the anomeric carbon was determined by J_{H1-H2} coupling constant and compared with published results.[15] As expected, glucose-based conjugated products (Lac, Mal, Glc, and GlcNAc) were formed

Table 1 Glycoconjugation of auxiliary **1** with reducing sugars



		Equivalent					
Entry	Reducing sugars	Reducing sugar (0.1 M)	**1**	Time (d)	Yield[a] (%)	Product	α-pyr : β-pyr : α-fur : β-fur[c]
1	D-Lactose (Lac)	5	1	1	64	**5**	0 : 100 : 0 : 0
2	D-Lactose (Lac)	1	5	2	51	**5**	0 : 100 : 0 : 0
3	D-Maltose (Mal)	5	1	2	51	**6**	0 : 100 : 0 : 0
4	D-Glucose (Glc)	5	1	2	66	**7**	0 : 100 : 0 : 0
5	*N*-Acetyl-D-glucosamine (GlcNAc)	5	1	3	46	**8**	0 : 100 : 0 : 0
6	D-Galactose (Gal)	5	1	2	67	**9**	0 : 85 : 0 : 15
7	D-Mannose (Man)	5	1	2	66	**10**	45 : 40 : 15 : 0[b]
8	D-Xylose (Xyl)	5	1	2	68	**11**	3 : 97 : 0 : 0

[a] Isolated yield of all isomers. [b] Determined by ^{1}H NMR. [c] Pyr = pyranose; Fur = furanose.

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exclusively in β-pyranose configuration. The xylose product was determined to also be predominately the β-pyranoside along with only trace amounts of α-pyranoside. The conjugation of galactose gave only β configured products, but was observed in both the pyranose and furanose forms in a ratio of 85 : 15. The exclusive formation of β-pyranoside can be rationalized by the favorable ring closing on the *Re* face[15a] in order to avoid a 1,3-axial interaction.[8f,15] The reaction of mannose with the tetrazine auxillary resulted in the most complex mixture due to the presence of axial alcohol on C2. Three different isomers including α-pyranoside, β-pyranoside, and α-furanoside were observed and difficult to distinguish. In contrast to glucose and galactose, the ring closing of mannoside occurs preferentially on the *Si* face and due to favourability of the opposing dipole–dipole interactions at C1 and C2 leading to the α-pyranoside as a major product.[8f,15]

Next, the hydrolytic stability of glycoconjugate **5** was studied in aqueous sodium phosphate buffer at pH values of 5, 7, and 9 at 40 °C. The reactions were monitored by HPLC and the concentration of glycoconjugate **5** was calculated from the absorbance at 254 nm (Fig. 1). The hydrolysis rate constants were calculated from the pseudo-first-order condition[16] (see ESI†). As expected, the hydrolysis rate of glycoconjugate **5** was most rapid at pH 5 with a half-life of 44 h. At pH 7, glycoconjugate **5** was very stable with half-life estimated to be 990 h providing a useful tool for the *in vitro* and *in vivo* investigation of glycan functions. Under basic conditions, the neoglycoside hydrolysis product was not observed. Instead, it was found that, at pH 9, the hydrolysis of the tetrazine moiety occurred with hydrazide (**12**), oxadiazole (**13**), and benzonitrile (**14**) derivatives observed as the degradation products[17] (Scheme 2).

With the intention of using the multifunctional auxiliary in applications beyond glycomics research, the compatibility and stability of auxiliary **1** was evaluated in the chemoenzymatic synthesis of complex oligosaccharides. The glycoconjugate **5**

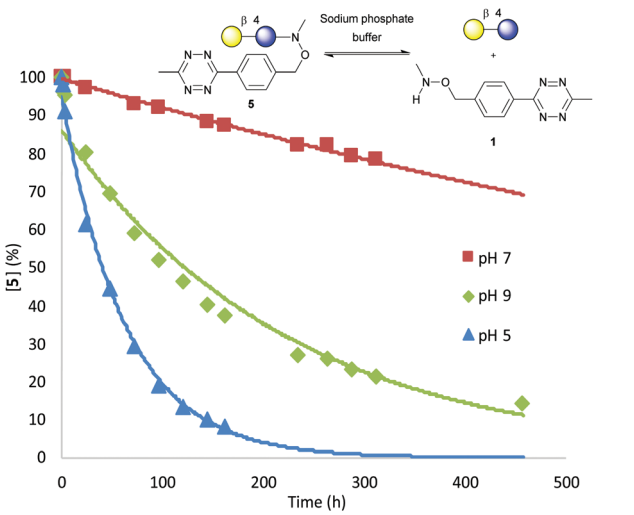
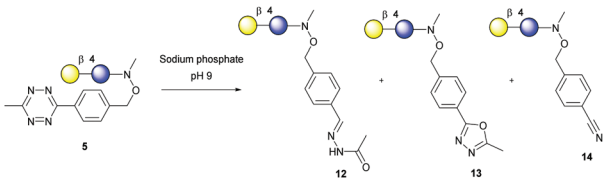
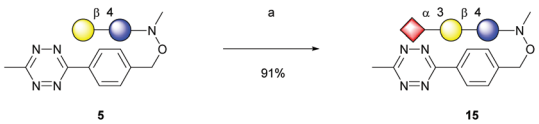


Fig. 1 Hydrolytic stability of glycoconjugate **5** at pH 5, 7, and 9 at 40 °C. Glycan symbol: ○ = D-galactose; ● = D-glucose.



Scheme 2 Degradation of glycoconjugate **5** in sodium phosphate pH 9. Glycan symbol: ○ = D-galactose; ● = D-glucose.

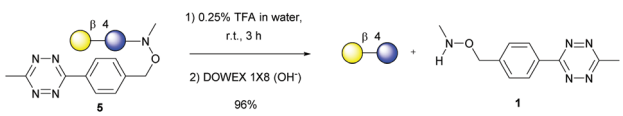


Scheme 3 Chemoenzymatic extension of glycoconjugate **5**: (a) α-(2 → 3)-sialyltransfe-rase, CMP-*N*-acetyl-neuraminic acid, Tris·HCl pH 8, 37 °C, 18 h. Glycan symbol: ○ = D-galactose; ● = D-glucose; ◇ = *N*-acetyl neuraminic acid.

was chosen to illustrate chemoenzymatic extension due to lactose being a common motif in human milk oligosaccharides.[8g,18] The glycoconjugate **5** and CMP-*N*-acetyl-neuraminic acid were treated with α-(2,3)-sialyltransferase resulting in the formation of trisaccharide **15** with the formation of α-(2,3)-Neu5Ac linkage to the lactose (Scheme 3). This auxiliary demonstrated tremendous utility by increasing the affinity of the trisaccharide product to a hydrophobic stationary phase during reverse phase chromatography relative to the unconjugated trisaccharide and could be observed by a standard UV detector. After completion, the product was isolated in excellent yield and purity after convenient purification on a preparative C18 column.

Another advantage of an oxyamine auxiliary is that it can be easily removed from the carbohydrate in the presence of acid. Previous studies have demonstrated that the oxyamine auxiliary can be selectively hydrolyzed with TFA without disturbing acid labile glycosidic linkages.[8d] Thus, glycoconjugate **5** was treated with 0.25% of TFA in water and stirred at room temperature for 3 h. The reaction was neutralized by DOWEX 1X8 (OH^- form) resin which could be easily removed by filtration (Scheme 4). Subsequently, the tetrazine auxiliary was removed by liquid–liquid extraction with dichloromethane and the unconjugated carbohydrate was recovered from the aqueous layer. After evaporation, the unconjugated lactose was obtained in 96% yield.

The structural complexity and high polarity of oligosaccharides in nature impede glycomics investigations due to the



Scheme 4 Removal of auxiliary from glycoconjugate **5**. Glycan symbol: ○ = D-galactose; ● = D-glucose.

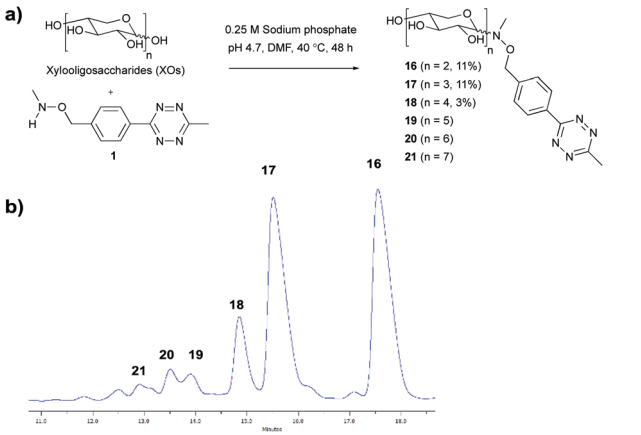
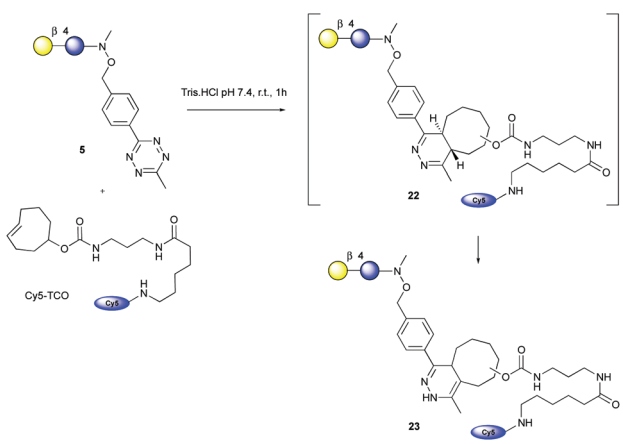


Fig. 2 Glycoconjugation and fractionation of XOs: (a) schematic conjugation reaction and (b) HPLC preparative chromatogram.

difficulty in purification and isolation. In order to demonstrate the utility of auxiliary **1**, the fractionation of a complex oligosaccharide mixture was attempted. A mixture of xylooligosaccharides (XOs) was chosen as it contained xylose oligomers of several chain lengths. The derivatization of the XOs and auxiliary **1** was carried out as previously described (Fig. 2a) and the reaction was monitored by HPLC. Subsequently, the glycoconjugates were purified by preparative reverse-phase chromatography. A HPLC method was identified to fractionate conjugated XOs up to 7 xylose units. Further method optimization would be possible for fractionation of longer glycans if they were present in higher abundance than the sample investigated in Fig. 2b. The conjugated xylodioside (**16**), xylotrioside (**17**), and xylotetroside (**18**) were collected and characterized by NMR and high-resolution mass spectrometry while the longer conjugated glycans were identified by low-resolution ESI mass spectrometry.

Finally, the ability to visualize tetrazine-functionalized oligosaccharides was demonstrated under biologically relevant conditions. Tetrazine rings, such as those in auxiliary **1**, can be readily coupled to chemical probes containing strained alkenes through an IEDDA reaction. The bioorthogonal labeling of glycoconjugate **5** with Cy5-TCO was chosen to be an example of the utility of tetrazine auxiliary in direct conjugation to chemical probe. The reaction was conducted by incubating the substrates in Tris buffer pH 7.4 at room temperature (Scheme 5). The reaction was monitored by HPLC using a biphenyl-functionalized silica column and UV detection at 647 nm which was specific for the Cy5. As expected, the starting materials were rapidly consumed to form an initial adduct (**22**), and then slowly rearranged to form a 1,4-dihydropyrazine **23**.[19] This rearrangement could be observed from HPLC chromatograms which proceeded to the completion within 18 h (see ESI†). The cycloaddition product **23** was then desalted using size-exclusion media (Sephadex™ G10) and characterized using mass spectrometry.



Scheme 5 IEDDA cycloaddition of glycoconjugate **5** and Cy5-TCO. Glycan symbol: ○ = D-galactose; ● = D-glucose.

Conclusions

In conclusion, a novel multifunctional tetrazine-containing neoglycoside auxiliary was successfully synthesized. The tetrazine auxiliary was conjugated with a variety of reducing sugars to create a tetrazine-glycan library. The hydrolytic stability was studied at various pH values revealing high stability at physiologically relevant conditions. The conjugated auxiliary could be efficiently cleaved with acid and conveniently removed by simple liquid–liquid extraction. The tetrazine auxiliary demonstrated usefulness by facilitating chemoenzymatic glycosylation and fractionation of complex oligosaccharides mixtures. Furthermore, the tetrazine moiety on the auxiliary allows for bioorthogonal labeling *via* IEDDA reaction and provides access to fluorescent visualization.

Conflicts of interest

There are no conflicts to declare.

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